

Capstone Timeline:

1. Create rough CAD model of the robot chassis using the dimensions listed on aliexpress, need to have a base with motor cutouts and can be disassembled for maintenance and parts access.
https://www.aliexpress.com/item/1005010074844733.html?spm=a2g0o.order_list.order_list_main.5.71521802pOgHsC
https://www.aliexpress.com/item/1005007016617326.html?spm=a2g0o.order_list.order_list_main.11.71521802pOgHsC
https://www.aliexpress.com/item/1005006500898019.html?spm=a2g0o.order_list.order_list_main.17.71521802pOgHsC
https://www.aliexpress.com/item/1005007792825403.html?spm=a2g0o.order_list.order_list_main.23.71521802pOgHsC
https://www.aliexpress.com/item/1005008449725840.html?spm=a2g0o.order_list.order_list_main.29.71521802pOgHsC
<https://pip-assets.raspberrypi.com/categories/892-raspberry-pi-5/documents/RP-008347-DS-1-raspberry-pi-5-mechanical-drawing.pdf?disposition=inline>
2. After parts have been delivered, test components to see compatibility, test current/voltage draw of all components and purchase associated batteries.
3. Assemble testing of all components and battery
4. Develop python programming for robot movement, camera movement, flash and operation.
5. Develop application IOS/Android for user interface with the robot with live streaming
6. Add LIPO battery charger to the robot
7. Create a sealed chassis for protection and must be water resistant.

GrabCAD models:

Wheels: <https://grabcad.com/library/85mm-wheel-1>

Motor driver: <https://grabcad.com/library/tb6612fng-driver-1>

Buck converter: <https://grabcad.com/library/ltc3780-step-up-and-step-down-regulator-module-1>

Camera: https://grabcad.com/library/usb-camera-module-1/details?folder_id=9352305

Raspberry Pi 5: https://grabcad.com/library/raspberry-pi-5-2/details?folder_id=13858786

Drive motors: <https://grabcad.com/library/jgb37-520-1>